Agent-Based Simulation of Organisational Misconduct

Reporting, Enforcement, and Hidden Misconduct

Final Presentation of Group 5
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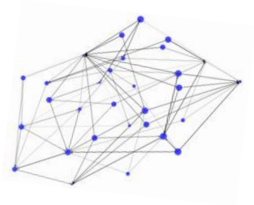


1. Problem setting

43%

of occupational fraud cases are detected through tips
ACFE, 2024

46%

of reported cases are confirmed as real misconduct after investigation
NAVEX, 2025

54%

of people who reported wrongdoing said they felt pressure not to report
EY, 2024

Organizational misconduct

Eg. financial fraud, expense fraud, hidden mistakes, bribery, etc...

- Organizations rely on **internal reporting** to detect misconduct
- **Enforcement policies** affect both misconduct and reporting behaviour

Key tension

- Stronger enforcement → **deterrence** but also **fear of retaliation**
- Reporting decisions are shaped by **fear of retaliation** and **reporter protection**.
- Fear can **reduce reporting** and thereby increase **hidden misconduct**

2. Research Question

Research Question

How do parameters such as punishment severity and reporter protection affect misconduct dynamics?

We are interested in the gap between:

Visible / punished misconduct and total misconduct in the organisation

Output	Meaning
Commited misconduct	Total amount of misconduct occurring in the organization
Punished misconduct	Misconduct that was observed, reported, and automatically punished
Hidden misconduct	Unreported or unnoticed misconduct
Hidden Misconduct Rate	**Captures the key question**: are policies reducing actual misconduct or simply increasing the share of hidden misconduct? = (Commited Misconduct - Punished Misconduct) / Commited Misconduct

3. Conceptual model

Agents

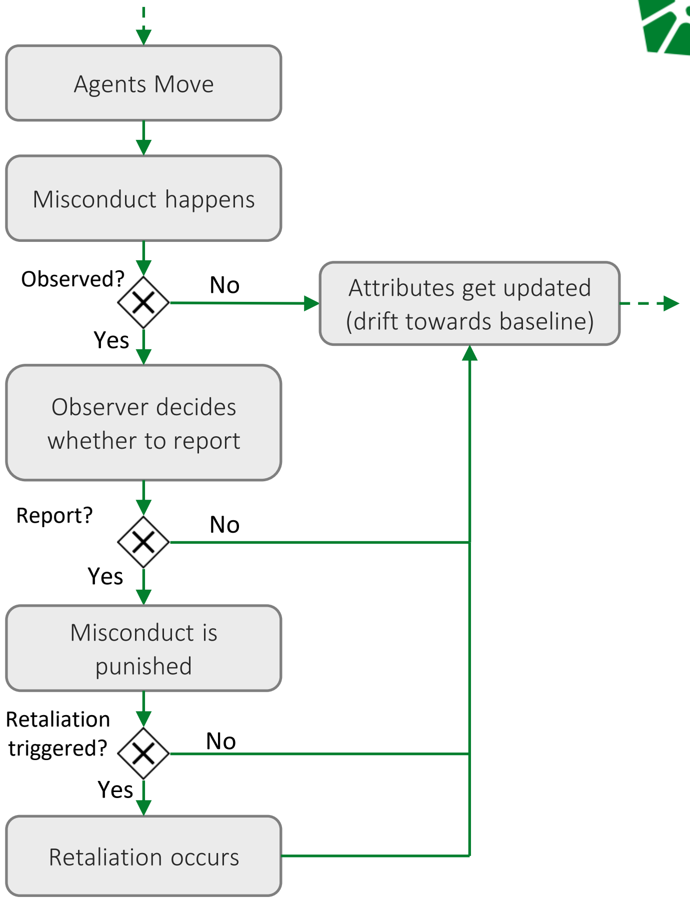
One Agent Type:
Employees

They can both
- Commit misconduct
- Report misconduct

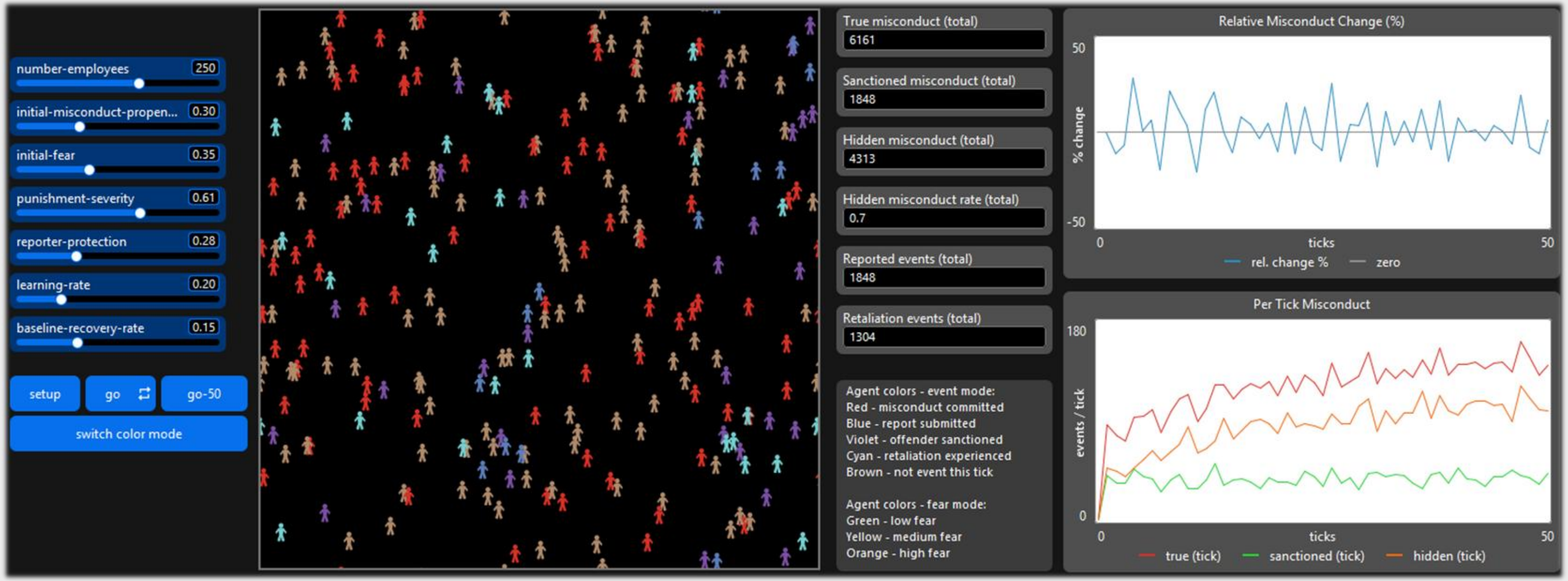
Environment

One room with a fixed population of employees

- Agents move randomly and interact locally
- Misconduct can only be reported if observed by nearby agents
- Punishment severity and reporter protection are determined globally
- Misconduct propensity and fear evolve over time through learning and feedback mechanisms



4. Demo of our simulation model



5. Experiment – How punishment and protection influence misconduct

Aspect	Description	
Purpose	Assess how punishment-severity and reporter-protection jointly affect committed misconduct, punished misconduct, hidden misconduct, retaliation, and fear.	
Fixed Parameters	number-employees = 300 initial-misconduct-propensity = 0.40 initial-fear = 0.30	learning-rate = 0.20 baseline-recovery-rate = 0.05 ticks = 300
Varied	Punishment-severity: 0.20–0.80 (step 0.05) Reporter-protection: 0.20–0.80 (step 0.05)	
Replications	10 per parameter combination	
Outcomes	Committed Misconduct Rate Punished Misconduct Rate Hidden Misconduct Rate	Retaliation Rate Mean Fear

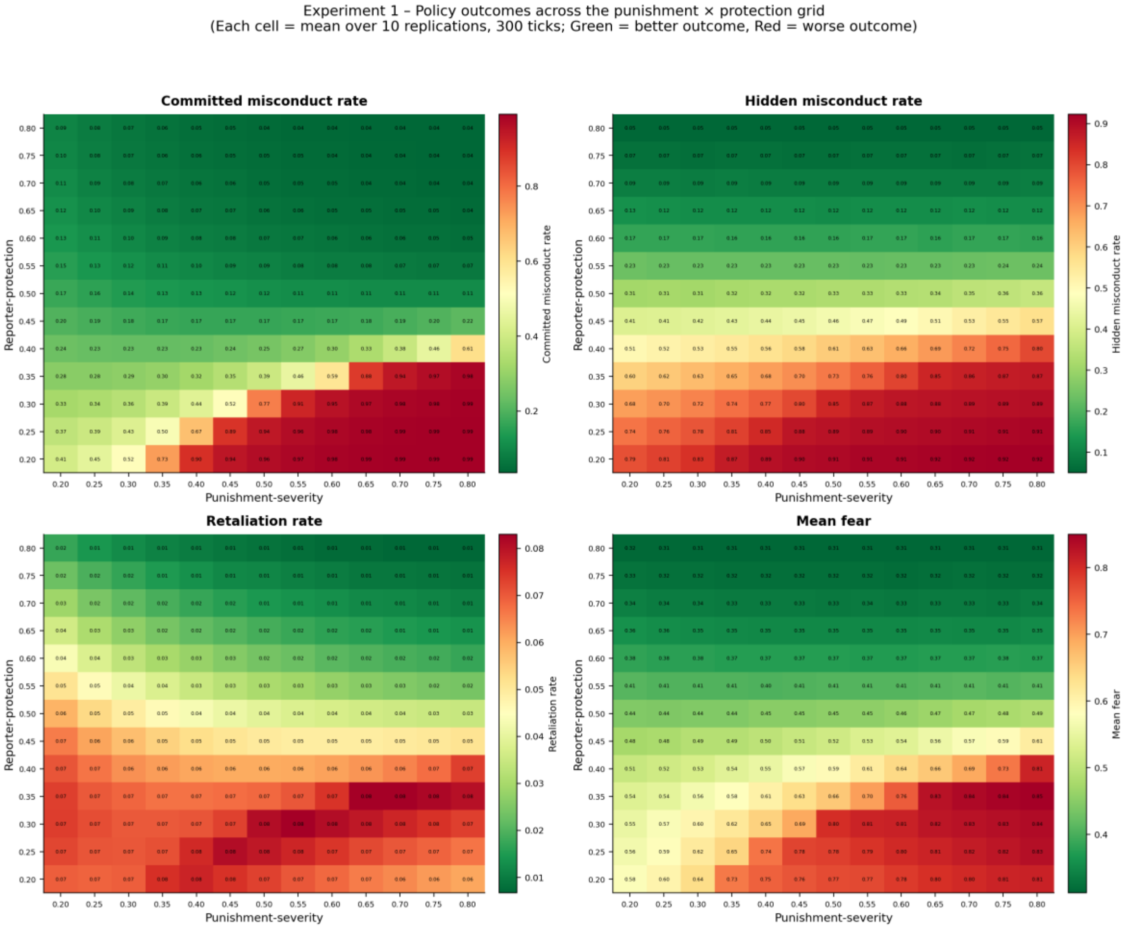
5.1 Experiment – Heatmaps

Findings

- Reporter protection reduces committed misconduct across all punishment severities
- Punishment severity alone has mixed effect
- Punishment severity with low reporter protection backfires, increasing hidden misconduct and fear

Answer to our research question

- Reporter protection is the only lever that reduces all four outcomes simultaneously
- Punishment severity only delivers deterrence when reporter protection is already high



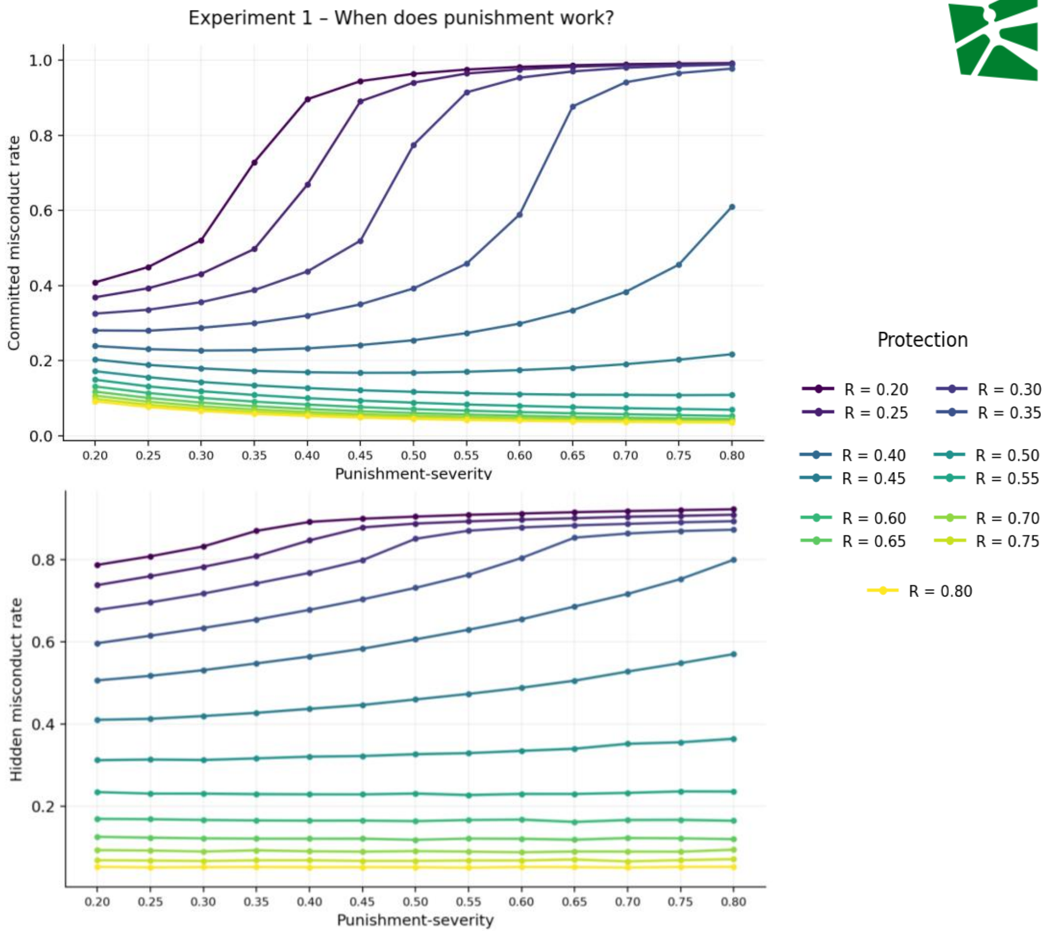
5.2 Experiment – Mechanism Plots

Findings

- Low reporter protection increases committed misconduct
- Increasing punishment severity can drive misconduct underground

Answer to our research question

- Punishment severity reduces committed misconduct only when reporter protection combats fear
- Otherwise, a “**chilling effect**” takes place



6. Validity of our model

Microvalidation

Making sense of our modelling choices

Agent rules are grounded in misconduct research:

- **Reporting decision** combines protection and fear: drivers from a whistleblowing meta-study (Mesmer-Magnus & Viswesvaran, 2005)
- **Retaliation = (1 – reporter protection)**: formal protection mechanically reduces successful retaliation
- **Different radii for misconduct observation and punishment witnessing**: represents the hidden nature of misconduct and the signal function of punishment

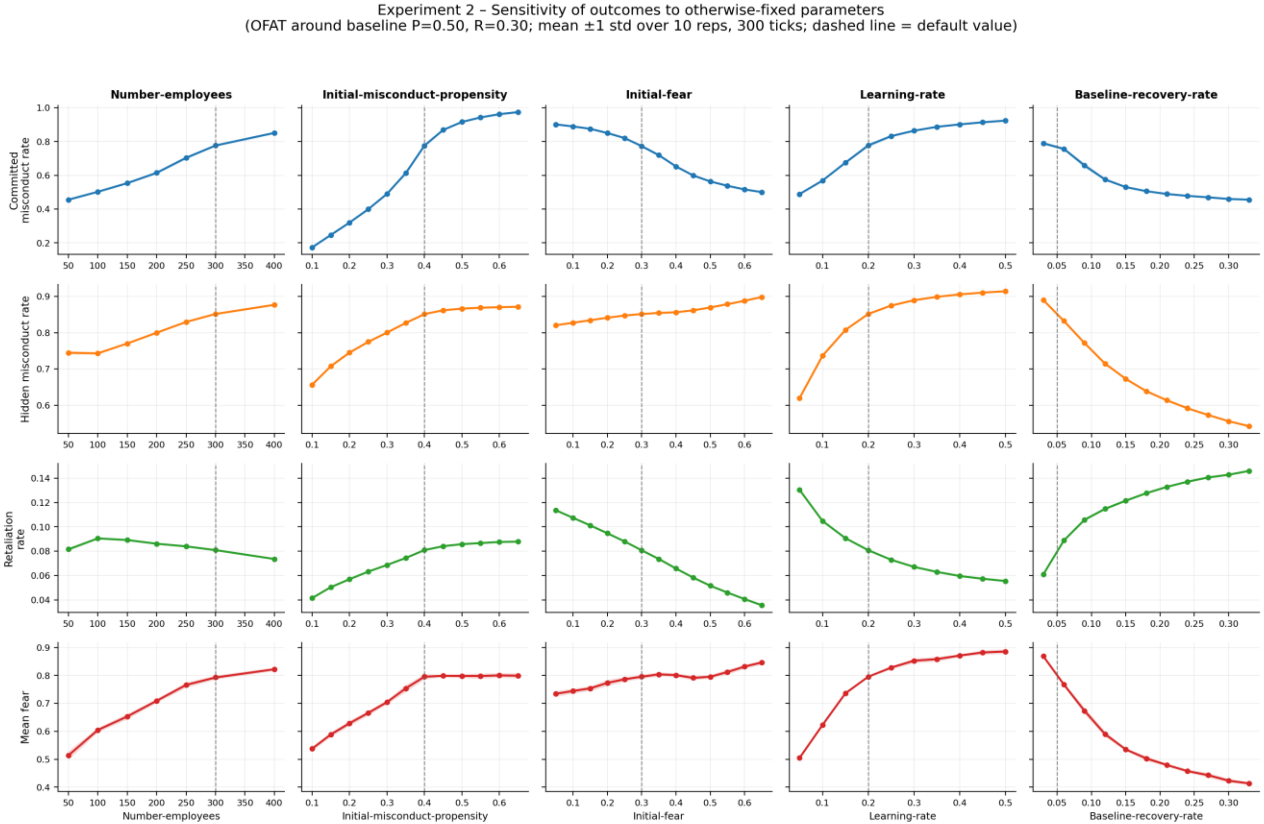
Macrovalidation

Making sense of our model observations

Emergent dynamics align with patterns in research:

- **Hidden vs. reported gap:** a lot of observed wrongdoing goes unreported (Near & Miceli, 1985)
- **Chilling effect**: protection ↓ → retaliation ↑ → fear ↑ → reports ↓ (Mesmer-Magnus & Viswesvaran, 2005)
- **Punishment without protection** shifts misconduct underground: corruption embeds and is concealed (Ashforth & Anand, 2003)

7. Validity of our model – Sensitivity Analysis



Main Findings

- **Tight ±1 std bands** show that stochastic noise is small and findings robust
- **Learning rate** influences the hidden misconduct rate, showing the fear trap / chilling effect
- **Baseline recovery** strongly affects the outputs

8. Personal Takeaways

Phenomenon

Insights about organisational misconduct

1. Reported ≠ committed misconduct – the management view can improve while misconduct persists
2. Emergence of the fear trap / chilling effect from local rules
3. The policies interact non-linearly – punishment severity and reporter protection are complements
4. Non-policy parameters influence the outcomes strongly

Methodological

Insights about agent-based modelling

1. Coding the model exposed gaps in our verbal description
2. Building the model was an iterative process
3. Level of detail depends on both realism and comprehensibility
4. Replaced a utility function with a learning-rate for simplification purposes

Thank you

Questions and feedback

Sources

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Appendix

A. Agent Attributes

Attribute		Meaning
Misconduct propensity	M	Likelihood that an employee engages in misconduct for personal benefit at any point in time. Decreases after punishment and adapts over time.
Fear of retaliation	F	Determines willingness to report misconduct. Increases after retaliation and decays gradually.
Location	L	Current location of the Agent. Determines whether misconduct can be observed by nearby agents.

B. Environmental Parameters (independent variables)

Parameter		Meaning
Number of employees	N	Determines interaction density and probability of observation
Initial misconduct propensity	M_0	Baseline tendency of agents to engage in misconduct
Initial fear	F_0	Initial reluctance to report misconduct
Punishment severity	P	Strength of sanctioning effect (enforcement intensity)
Reporter protection	R	Reduces retaliation risk and increases reporting probability
Learning rate	λ	Speed at which agents adapt behaviour over time
Baseline recovery rate	β	Speed at which misconduct propensity and fear revert toward their initial values when no shock occurs in a given tick

C. OLS Regression Results

Variables

Dependent:

- Hidden misconduct rate

Independent:

- Punishment severity
- Reporter protection

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                            OLS Regression Results
==============================================================================
Dep. Variable:     hidden-misconduct-rate   R-squared:                   0.945
Model:                                OLS   Adj. R-squared:              0.945
Method:                     Least Squares   F-statistic:                 9643.
Date:                    Thu, 07 May 2026   Prob (F-statistic):           0.00
Time:                            10:39:20   Log-Likelihood:             1993.1
No. Observations:                    1690   AIC:                        -3978.
Df Residuals:                        1686   BIC:                        -3956.
Df Model:                               3
Covariance Type:                nonrobust
=======================================================================================
                          coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------
const                   0.9522      0.015     64.536      0.000       0.923       0.981
punishment-severity     0.5649      0.028     20.440      0.000       0.511       0.619
reporter-protection    -1.2399      0.028    -44.864      0.000      -1.294      -1.186
interaction            -0.7833      0.052    -15.131      0.000      -0.885      -0.682
==============================================================================
Omnibus:                      260.699   Durbin-Watson:                   0.235
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               63.789
Skew:                           0.106   Prob(JB):                     1.41e-14
Kurtosis:                       2.072   Cond. No.                         45.2
==============================================================================
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Stronger reporter protection consistently reduces hidden misconduct, while punishment alone (without adequate protection) increases fear, suppresses reporting, and drives more misconduct underground.

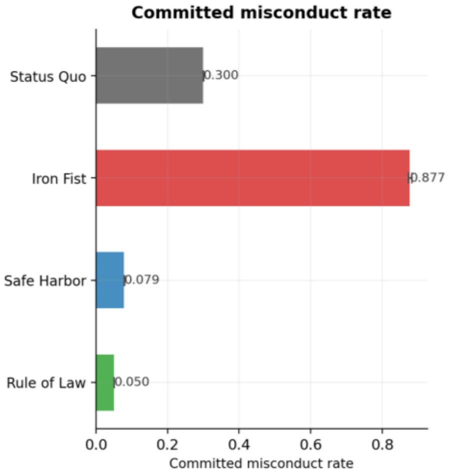
D. Experiment – Archetypes

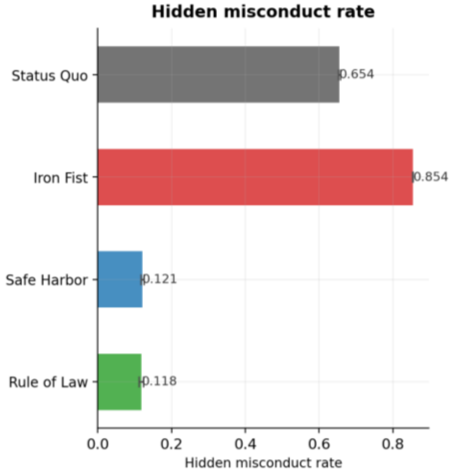
Four archetypes (Punishment, Protection)

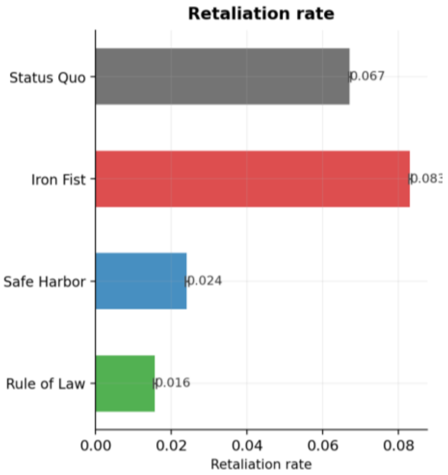
- Status Quo (0.35,0.35)
- Iron Fist (0.65,0.35)
- Safe Harbor (0.35,0.65)
- Rule of Law (0.65,0.65)

Takeaway

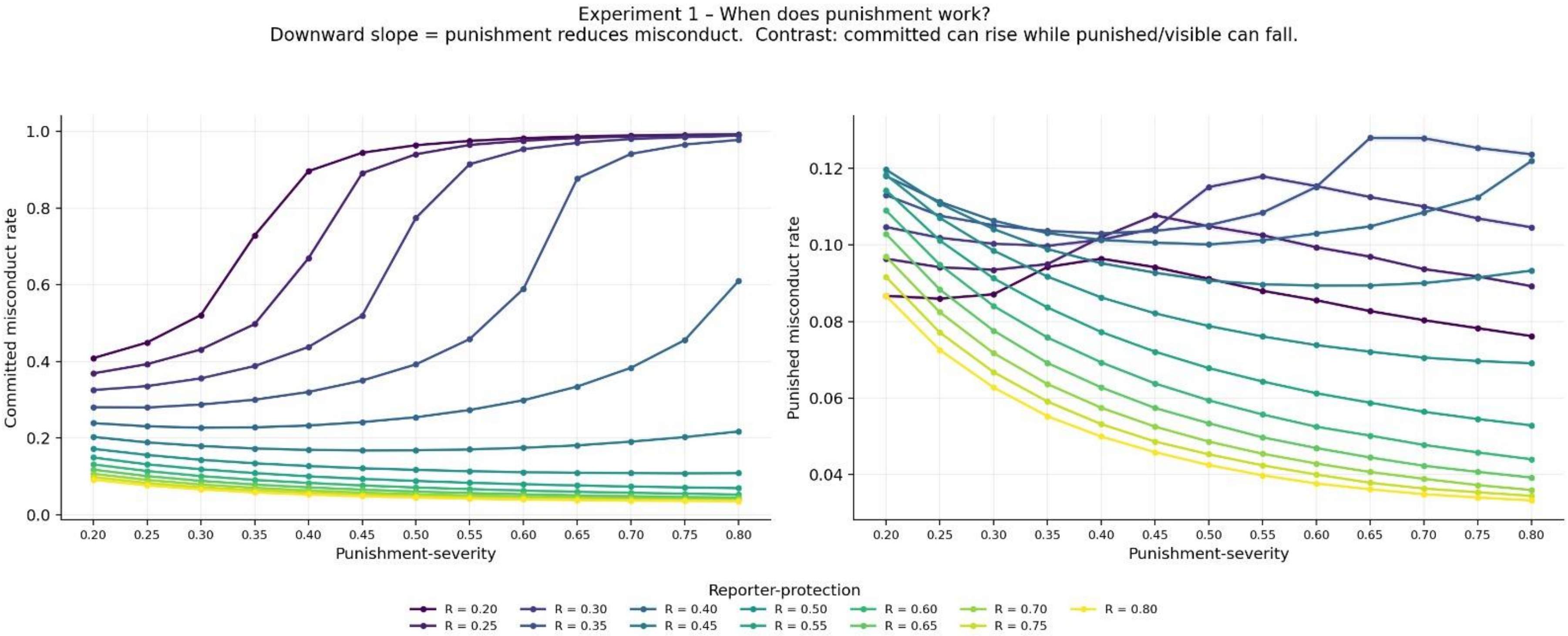
- Punishment and protection are complements
- Without protection misconduct will be hidden
- Combining both is essential







E. Experiment — Punished misconduct rate



F. Experiment — Trade-offs